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WO-400-04 — Remake flanged joint 3"-P300-P-1015/B12 on 3"-P300-P-1015

Area	A400 Pumps	**Task**	GJ Flanged joint remake
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	P-103A — P-103A centrifugal pump — E 174'-0" N -18'-6" EL 100'-9"		
Equipment	P-103A	**Lines**	3"-P300-P-1015
Isometrics	ISO-3"-P300-P-1015 (generated); owner's drawings ISO-P-1015		
Permits	General work permit		

Reason

Weeping joint at the control valve station.

Scope of work

1. Isolate per the isolation list, drain and vent 3"-P300-P-1015.
2. Break the joint at E 68'-6" N 17'-8-13/16" EL 125'-0" between VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300 and VALVE, GATE, WEDGE, OS&Y 3" P300.
3. Inspect both flange faces (SW-NDE-5).
4. Fit a new gasket and new studs; tighten to the stated preload in three passes, cross pattern (SW-WO-5).
5. Remove blinds; leak test on start-up (SW-TEST-2).

Affected items: 3"-P300-P-1015/B12, 3"-P300-P-1015/B13, 3"-P300-P-1015/B10

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1015	3"-P300-P-1015	3"-P300-P-1015	6	1 error(s), 9 warning(s) — see below

ISO-P-1015 (PDF): ../isometrics-imported/ISO-P-1015.pdf — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1015	1	error	sheet 1: CONT'D ON ISO-P-1016 prints E 174'-2 1/2" N -17'-6" EL 107'-6" for E 174'-2 1/2" N -16'-6" EL 107'-6": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 3"-P300-P-1015 meets 3"-P300-P-1016 (E 174'-2 1/2" N -16'-6" EL 107'-6")
ISO-P-1015	5	warning	weld 3"-P300-P-1015/W16 (E 68'-6" N -5'-0" EL 124'-7", sheet 5) has no label
ISO-P-1015	5	warning	weld 3"-P300-P-1015/W17 (E 68'-11" N -5'-0" EL 125'-0", sheet 5) has no label
ISO-P-1015	5	warning	weld 3"-P300-P-1015/W19 (E 69'-6" N -4'-7" EL 125'-0", sheet 5) has no label
ISO-P-1015	5	warning	weld 3"-P300-P-1015/W20 (E 69'-6" N -4'-2 1/2" EL 125'-0", sheet 5) has no label
ISO-P-1015	6	warning	W22 (3"-P300-P-1015/W22, sheet 6) is labelled again on sheet 5; it is 149.8 in from the nearest break
ISO-P-1015	6	warning	W23 (3"-P300-P-1015/W23, sheet 6) is labelled again on sheet 5; it is 159.8 in from the nearest break
ISO-P-1015	6	warning	weld 3"-P300-P-1015/W26 (E 68'-6" N 14'-11 9/16" EL 125'-0", sheet 6) has no label
ISO-P-1015	5	warning	branch weld 3/4"-P300-I-7015/W1 (E 68'-8 1/2" N -5'-0" EL 124'-0", sheet 5) has no label
ISO-P-1015	5	warning	branch weld 3/4"-P300-SP-7026/W1 (E 69'-3 1/2" N -4'-0" EL 125'-0", sheet 5) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-103A	equipment	P-103A centrifugal pump		E 174'-0" N -18'-6" EL 100'-9"
3"-P300-P-1015/B12	item	GASKET 3" P300 COMPRESSED SHEET	3"-P300-P-1015	E 68'-6" N 17'-8-13/16" EL 125'-0"
3"-P300-P-1015/B13	item	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300	3"-P300-P-1015	E 68'-6" N 17'-1-5/8" EL 125'-0"
3"-P300-P-1015/B10	item	VALVE, GATE, WEDGE, OS&Y 3" P300	3"-P300-P-1015	E 68'-6" N 18'-3-11/16" EL 125'-0"
3"-P300-P-1015/B16	item	VALVE, GATE, WEDGE, OS&Y 3" P300	3"-P300-P-1015	E 68'-6" N 15'-11-9/16" EL 125'-0"
3"-P300-P-1015/B2	item	VALVE, GATE, WEDGE, OS&Y 3" P300	3"-P300-P-1015	E 68'-6" N 21'-9-3/4" EL 125'-0"
3"-P300-P-1015/B5	item	FLANGE WELD-NECK 3" P300 RAISED FACE	3"-P300-P-1015	E 68'-6" N 21'-1/4" EL 125'-0"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	3"-P300-P-1015/B16	3"-P300-P-1015	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (3"-P300-P-1015/B16)	E 68'-6" N 15'-11-9/16" EL 125'-0"		
2	blind	3"-P300-P-1015/B13	3"-P300-P-1015	Fit a line blind between 3"-P300-P-1015/B16 and VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300 (3"-P300-P-1015/B13), work side	E 68'-6" N 17'-8-3/4" EL 125'-0"		
3	valve	3"-P300-P-1015/B2	3"-P300-P-1015	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (3"-P300-P-1015/B2)	E 68'-6" N 21'-9-3/4" EL 125'-0"		
4	blind	3"-P300-P-1015/B5	3"-P300-P-1015	Fit a line blind between 3"-P300-P-1015/B2 and FLANGE WELD-NECK 3" P300 RAISED FACE (3"-P300-P-1015/B5), work side	E 68'-6" N 21'-2-13/16" EL 125'-0"		

Parts

Description	Qty	Replaces
GASKET 3" P300 COMPRESSED SHEET	1	3"-P300-P-1015/B12
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	3"-P300-P-1015/B14

Inspections

Id	Inspection	Criteria	Basis
WO-400-04-I1	Flange faces before the new gasket goes in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
3"-P300-P-1015/B12	4 × 0.875" studs (3"-P300-P-1015/B14)	27,059	3	

Test — Service leak test

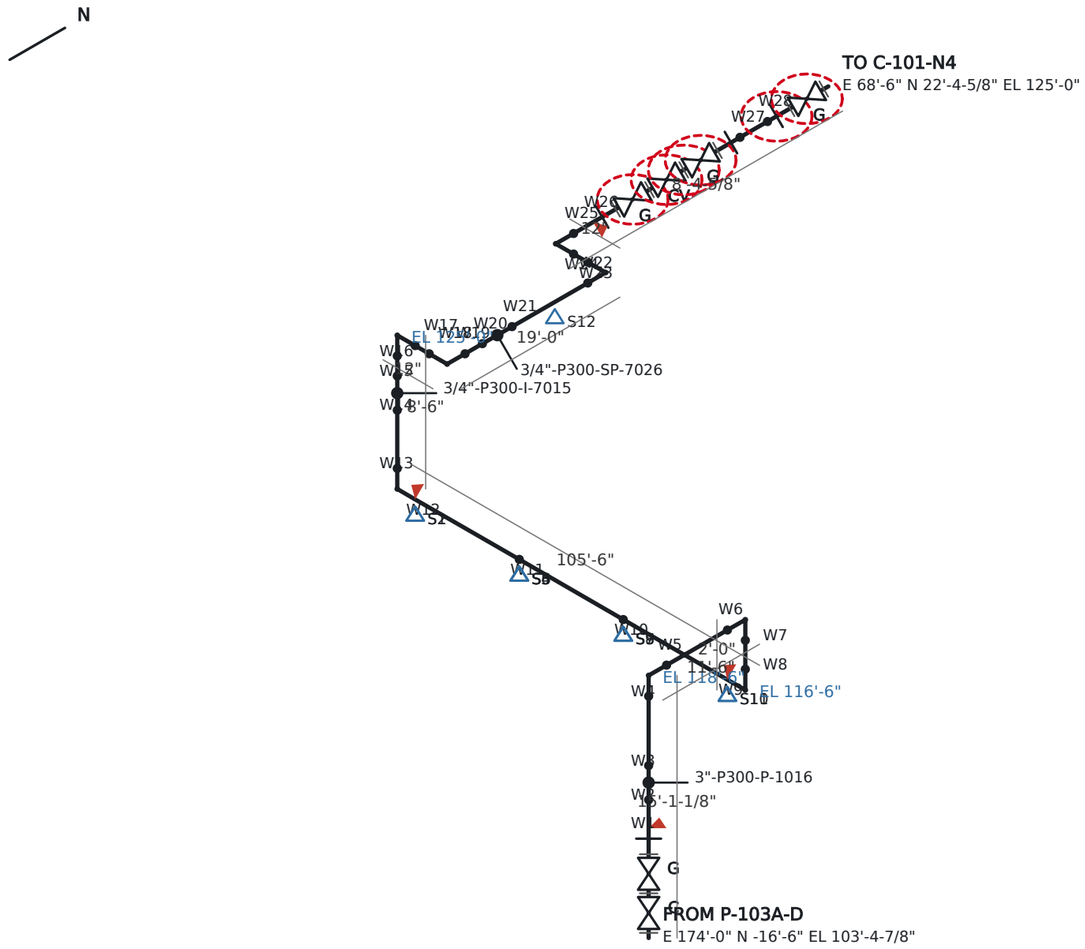
Description	The remade joint checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Joint broken; both flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New gasket fitted; studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated	Bolt-up record	W	W	—				
4	Service leak test on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
5	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
6	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 3"-P300-P-1015 — Remake flanged joint 3"-P300-P-1015/B12 on 3"-P300-P-1015



BILL OF MATERIALS

1	GASKET 3" P300 COMPRESSED SHEET	9
2	VALVE, SWING CHECK 3" P300	1
3	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	9
4	VALVE, GATE, WEDGE, OS&Y 3" P300	4
5	FLANGE WELD-NECK 3" P300 RAISED FACE	4
6	PIPE 3" P300 BEVELLED ENDS	155'-4-3/8"
7	TEE EQUAL 3" P300	1
8	ELBOW 90° R = 5/3 OD 3" P300	8
9	TEE REDUCING 3" × 3/4" P300	2
10	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300	1

WELDS 28 (4 FIELD) · SUPPORTS 12 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9F W10
W11 W12F W13 W14 W15 W16 W17 W18 W19 W20
W21 W22 W23 W24 W25 W26F W27 W28

3"-P300-P-1015

CLASS H300 — Hydrocarbon process
SERVICE P · 3" · P300 · DESIGN 300 psi
P-103A-D → C-101-N4
PIPE 155'-4-3/8" · WEIGHT 1731 lb

DWG ISO-3"-P300-P-1015 REV 0 WO-400-04

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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